

Noise Thresholds for Jacobian-Based Embedding Criteria and a Residual-Whiteness Alternative

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Abstract

Delay embedding parameter selection relies on diagnostics — false nearest neighbours, Jacobian-trace convergence, prediction error — that implicitly assume deterministic, noise-free dynamics. We show empirically that these criteria share a noise threshold, above which they lose diagnostic value: for the Lorenz-63 system, false nearest neighbour percentages at the correct embedding dimension rise from 2% (clean) to 62% (5% noise), while the Jacobian trace becomes uninformative.

We propose residual autocorrelation convergence — the decrease of lag-1 autocorrelation in local-linear-fit residuals as embedding dimension increases — as a noise-robust alternative. This criterion correctly identifies sufficient embedding dimension under 5–20% observational noise where deterministic criteria fail, and provides a U-shaped minimum for lag selection. Results are validated on the Rössler system.

The computational primitive — local linear regression in delay coordinates — is shared with S-maps [10] and Jacobian-based Lyapunov estimation [8]. The contribution is the application to embedding assessment and the characterization of noise operating regimes.

Keywords. delay embedding, embedding dimension, noise robustness, local linear model, false nearest neighbours, residual diagnostics, time series analysis.

1 Introduction

Delay embedding [11, 9] reconstructs the state space of a dynamical system from a scalar time series $\{y_t\}$ via delay vectors

$$\mathbf{z}_t = (y_t, y_{t-L}, \dots, y_{t-(d-1)L}) \in \mathbb{R}^d. \quad (1)$$

The embedding dimension d and lag L must be chosen from data. Standard criteria include false nearest neighbours [5] (FNN), mutual information for lag selection [4], prediction error minimisation [7], and convergence of Lyapunov exponent estimates [8].

These criteria assume the underlying dynamics are deterministic and the observations are noise-free. Under observational noise, they degrade. FNN in particular is known to be sensitive to noise [7], but the degradation has not been systematically characterised.

We characterise the threshold, propose a noise-robust alternative, and explain why deterministic criteria share the same vulnerability.

2 Method

At each point $\mathbf{z}$ in the reconstructed space, we fit a local linear model

$$\mathbf{z}_{t+1} \approx A(\mathbf{z})(\mathbf{z}_t - \mathbf{z}) + \mathbf{b}(\mathbf{z}) \quad (2)$$

by weighted least squares with a compactly supported tricube kernel $w(u) = (1 - u^3)^3$ for $\|u\| \leq 1$. Neighbours within bandwidth h are identified via KD-tree. The Jacobian $A(\mathbf{z}) \in \mathbb{R}^{d \times d}$ is the local linearisation of the pushforward map.

This computation is shared with S-maps [10, 2] (used for prediction and causality detection) and Jacobian-based Lyapunov estimation [8, 3] (used to compute dynamical invariants). We apply it to a different question: embedding assessment.

From the local fit we extract two diagnostics:

Deterministic diagnostic: the Jacobian trace

$$\text{div } F(\mathbf{z}) = \text{tr}(A(\mathbf{z})), \quad (3)$$

averaged over the attractor. For a faithful embedding of a deterministic system, $\langle \text{tr}(A) \rangle$ converges to the sum of local Lyapunov exponents as d increases. Instability of $\langle \text{tr}(A) \rangle$ with d indicates insufficient embedding.

Residual diagnostic: the lag-1 autocorrelation of residuals

$$\varepsilon_t = \mathbf{z}_{t+1} - (A(\mathbf{z}_t)(\mathbf{z}_t - \mathbf{z}_{\text{nn}}) + \mathbf{b}(\mathbf{z}_{\text{nn}})), \quad (4)$$

where $\mathbf{z}_{\text{nn}}$ is the nearest query point. If the embedding is sufficient and the local linear model captures the deterministic dynamics, $\{\varepsilon_t\}$ should be white noise. Remaining autocorrelation indicates deterministic structure not captured by the embedding.

The criterion: *the embedding dimension is sufficient when the residual lag-1 autocorrelation stabilises near zero*. For lag selection: *the optimal lag minimises the residual autocorrelation at sufficient d* .

3 Noise threshold for deterministic criteria

Both diagnostics of §2 require a local linear fit, which in turn requires neighbours within bandwidth h . We test on Lorenz-63 ($N = 50\,000$, $\Delta t = 0.01$, $h = 5$, $L = 10$), observing the x -component corrupted by additive Gaussian noise at amplitudes 0–20% of the signal standard deviation.

Table 1 shows FNN percentages (threshold $R = 2$) at the correct embedding dimension $d = 3$ as a function of noise level.

Noise (%)	0	1	2	5	10	20
FNN at $d = 3$ (%)	1.8	27.0	40.6	62.4	74.5	79.4

Table 1: False nearest neighbour percentage at the correct embedding dimension $d = 3$ for the Lorenz-63 system under varying observational noise. Above $\sim 2\%$ noise, FNN incorrectly rejects the true dimension.

The mean $|\langle \text{tr}(A) \rangle|$ across the attractor shows a similar threshold. For clean data, it increases from 0.7 ($d = 2$) to 47.7 ($d = 6$), clearly indicating dimension convergence. At 5% noise, the range collapses to 0.02–0.68: no diagnostic signal remains.

The transition is sharp: at 2% noise, diagnostic signal persists (range 0.7–7.2); at 5%, it vanishes. Figure 1 shows the full noise–dimension phase diagram for both diagnostics.

4 Residual autocorrelation as embedding criterion

The Jacobian trace fails above $\sim 2\%$ noise because it tests a property (well-defined local expansion rate) that noise destroys. The residual diagnostic tests a different property: absence of predictable structure in the fit errors.

Table 2 shows the lag-1 residual autocorrelation as a function of embedding dimension, for both clean and 5%-noisy Lorenz data.

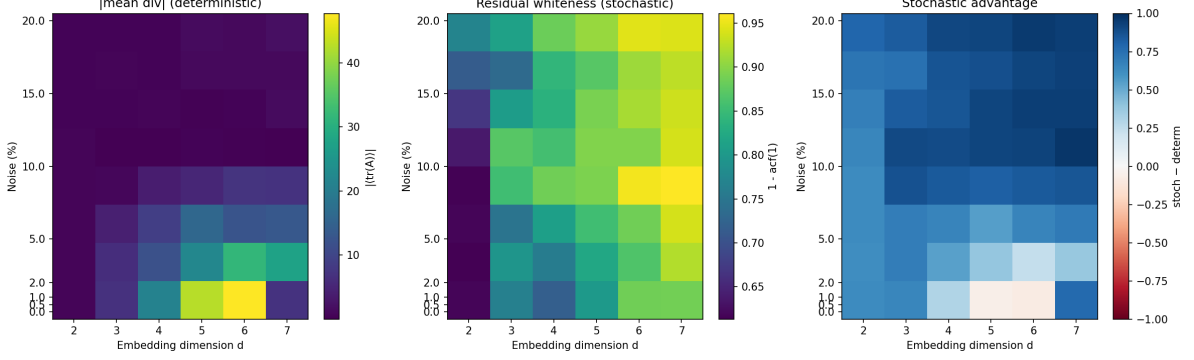


Figure 1: Phase diagram: noise level vs. embedding dimension for the deterministic diagnostic (left: $|\langle \text{tr}(A) \rangle|$) and the residual diagnostic (centre: $1 - \text{acf}(1)$, where higher = whiter residuals). The deterministic diagnostic loses signal above ~ 2 –5% noise; the residual diagnostic functions across the full range.

d	2	3	4	5	6	7
Clean: $\text{acf}(1)$	0.38	0.24	0.28	0.20	0.11	0.11
Noisy (5%): $\text{acf}(1)$	0.36	0.13	0.14	0.11	0.11	0.06

Table 2: Lag-1 residual autocorrelation versus embedding dimension. Both clean and noisy data show a decreasing trend, stabilising at $d \approx 5$ –6. The criterion works under noise where FNN and Jacobian trace do not.

The residual autocorrelation decreases with d and stabilises across the full noise range tested (0–20%).

At fixed $d = 3$, the residual autocorrelation as a function of lag L shows a U-shaped profile with a minimum near $L \approx 7$ –11 (corresponding to 0.07–0.11 orbital periods). This minimum persists under 5% noise (Figure 2), providing a noise-robust lag selection criterion.

The same patterns hold for the Rössler system ($a = 0.2$, $b = 0.2$, $c = 5.7$): residual autocorrelation decreases with d for both clean ($0.14 \rightarrow 0.03$) and noisy data ($0.09 \rightarrow 0.01$), while the Jacobian trace diagnostic works only for clean data.

5 Discussion

The shared noise vulnerability of FNN, Jacobian trace, and Lyapunov convergence is not coincidental. All three test for properties of deterministic dynamics: FNN tests for injectivity of the delay map; the Jacobian trace tests for well-defined local expansion rates; Lyapunov convergence tests for stability of dynamical invariants. Observational noise destroys injectivity (nearby points in delay space may be far apart in state space due to noise), corrupts Jacobian estimates, and introduces spurious Lyapunov exponents.

The residual criterion survives because it tests a different property: whether the local linear model has captured all *predictable* structure, regardless of noise. Residual autocorrelation measures remaining deterministic signal in the prediction errors — a property that is well-defined even when the dynamics are stochastic. The embedding is sufficient when no deterministic structure remains in the residuals.

This distinction corresponds to a difference in modelling commitment. Deterministic criteria implicitly assume $\mathbf{z}_{t+1} = F(\mathbf{z}_t)$ (single-valued dynamics); noise violates this assumption and the criteria respond to the violation rather than to embedding insufficiency. The residual criterion assumes $\mathbf{z}_{t+1} = F(\mathbf{z}_t) + \sigma(\mathbf{z}_t)\boldsymbol{\eta}_t$ (drift plus diffusion); noise is built into the model, and the

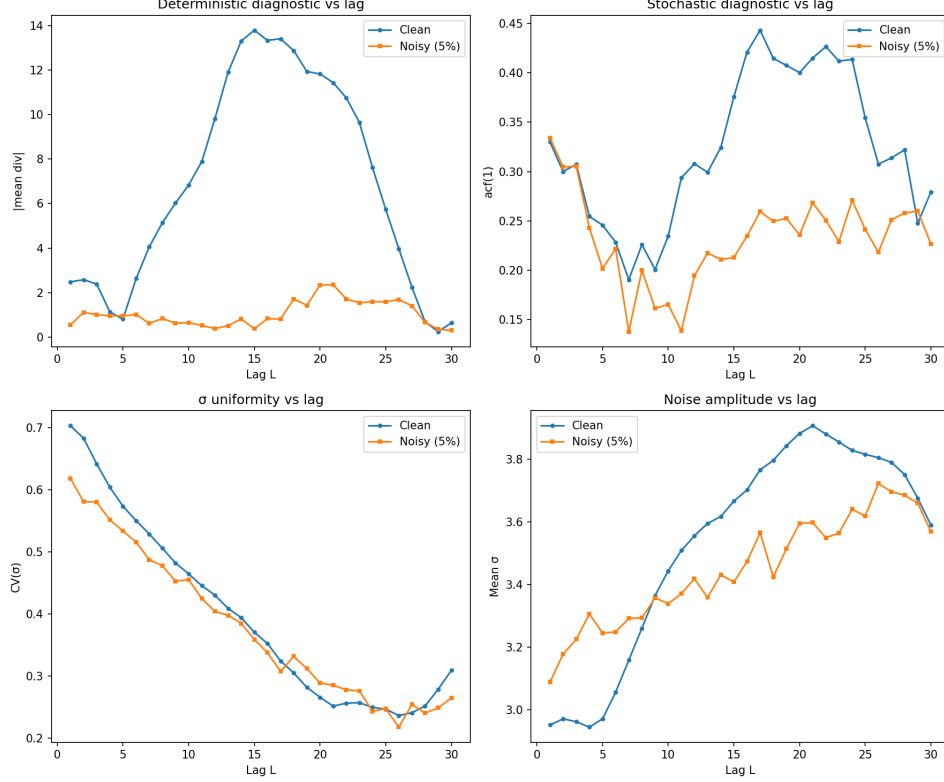


Figure 2: Lag sweep ($d = 3$, Lorenz-63). Top left: Jacobian trace vs. lag (bell-shaped for clean, flat for noisy). Top right: residual $\text{acf}(1)$ vs. lag (U-shaped for both clean and noisy, minimum at $L \approx 7-11$).

diagnostic tests only for remaining deterministic structure.

The embedding problem is algebraically underdetermined: the observation algebra determines a measure on compatible observation-patterns but does not constrain which patterns correspond to realised states [6]. This is why embedding selection is necessarily empirical, and why different empirical criteria have different operating regimes. The practical recommendation: for clean data ($< 2\%$ noise), deterministic criteria (FNN, Jacobian convergence) provide sharp diagnostics. Above this threshold, use residual autocorrelation convergence for dimension selection and the U-shaped minimum for lag selection.

The local linear computation is shared with S-maps [10, 2, 1] (prediction and causality) and Ragwitz–Kantz [7] (embedding selection via prediction error). Ragwitz–Kantz minimise out-of-sample prediction error over (d, L) , which is prediction-optimised; our residual autocorrelation criterion is sufficiency-based. These can disagree: prediction skill may plateau before residuals whiten (diagnostic underfitting) or overfit in noisy regimes.

Cenci et al. [1] regularise S-maps under noise; regularisation of the local Jacobian may extend the operating regime of the deterministic diagnostic, but this is not explored here.

Limitations

The noise threshold ($\sim 2-5\%$) is characterised empirically for two systems (Lorenz-63, Rössler). Its dependence on attractor dimension, data length, and bandwidth is not systematically explored. The residual criterion requires sufficient data to estimate local linear maps reliably; in very high dimensions or with sparse data, both criteria may fail.

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